Google Cloud H2S HACKATHON

AI Builder Cup

Build in JAPAC.
Build for the world.



Team Details

a. Team name: **Techno Crackers**

b. Team leader name: **Venket Raj S** (Team: Venket Raj S · Malik Chandra Pandian · Arun Veerappan · Nagaraj — 4 working professionals in JAPAC, no students)

c. Problem Statement: **BFSI: Intelligent Risk, Fraud & Financial Experiences**

CatSight — Event-aware contract intelligence for catastrophe reinsurance: After a typhoon or earthquake, a reinsurer cannot say quickly or defensibly how much it owes because the answer depends on contract clauses buried in PDF treaty wordings.

Brief about the idea

CatSight tells a reinsurer in **about 9–15 seconds** after a typhoon or earthquake which contracts respond, what each layer pays, and which contract clauses decide that payout — with every quote machine-verified against the source PDF page.

The Human & Community Problem:

- When a catastrophe hits, **primary insurance companies** must rapidly pay hundreds of thousands of claims to rebuilding families and local businesses (e.g., **857,284 claims & ¥1,067.8bn paid** after Typhoon Jebi (GIAJ)).
- To stay solvent, primary insurers rely on multi-million dollar payouts from **reinsurance contracts**.
- **The bottleneck:** Reinsurance contracts are 50-page legal PDFs. While industry loss models take **4–20 days**, claims teams spend **30–60 analyst-hours** manually reviewing contracts and arguing over ambiguous exclusions (like windstorm vs. storm-surge flooding). When reinsurance payouts stall, community recovery stalls.

The Plain-English Translation: *Primary Insurer (Cedent)* = company paying local claims; *Deductible (Retention)* = what the insurer pays first; *Coverage Used (Burn %)* = percentage of policy limit paid out; *Fee to Refill Coverage (RIP)* = fee to restore protection after a claim.

Source / Workflow	Time to Arrive	Scope & Practical Limitation
Cat-modeller event response	**4–20 days**	Broad market range; ignores your specific PDF contracts
PERILS (Asia loss index)	**6 weeks**	Industry loss index, not treaty-level payout calculation
Verisk PCS Japan	**up to 90 days**	Market-wide index for events ≥ $2bn
Manual in-house spreadsheet review	**2–4 days**	Slow, error-prone; missed endorsements and exclusions
CatSight (Lion Re's book)	**0.7 s math · ~9–12 s full AI check**	**Event × exposure × layer math × verified PDF wording**

Replays real public hazard footprints (NOAA IBTrACS, USGS ShakeMap) over a synthetic book of **24 reinsurance contracts written for 8 primary insurers** in Japan, the Philippines, and Taiwan.

Opportunities

- **How different is it from any of the other existing ideas?**

  • **CatSight is the join:** Cat-model event response (Moody's RMS, Verisk) estimates hazard footprints on generic terms, stopping before the contract PDF. Document-AI tools extract clauses at underwriting time, stopping before the storm track and financial math. Generic LLM chatbots over PDFs hallucinate arithmetic and lack an audit trail.
  • **Complementary by design:** CatSight is not a cat model; a licensed vendor footprint plugs directly into our typed hazard interface.

- **How will it be able to solve the problem?**

  • **Deterministic Python loss engine (`Decimal`, 874 offline tests + 13 opt-in live tests):** Computes regional gross loss, tier payouts, coverage limit used (burn %), fee to refill coverage (reinstatement premium), and net shares in **0.7–1.0 s** — the LLM never does arithmetic.
  • **RAG over Firestore vector search + 2-Stage Verifier Loop:** Retrieves governing clauses per contract (252 vectors, 768-d `gemini-embedding-001`) and verifies quotes via deterministic fragment matching + a Gemini 3.8 Flash semantic support check (with 1 revision retry before withholding).

- **USP & Social Impact of the proposed solution**

  • **Adversarial Clause Courtroom:** Two AI counsel debate opposing readings of ambiguous clauses (e.g., T-004 storm-surge exclusion) from verified quotes while the engine prices both readings, showing the exact **$14.6M disputed claim difference**.
  • **Scanned Endorsement Detective:** Transcribes stamped, rasterised PDF endorsements with no text layer (T-004 p.7: 7-day storm window replaces 3 days) and badges them `transcribed`.
  • **Community & APAC Protection Gap:** 91% of APAC's disaster losses ($65bn in 2023) were uninsured (MAS), and 48% of regional insurers license no cat model (Aon). Faster, transparent reinsurance settlements let primary insurers pay households and small businesses sooner.

List of features offered by the solution

Feature	What it gives the claims analyst & primary insurer	Verification & Trust Guardrail
1. Contract Matching & Payout Waterfall	Matches all 24 contracts on peril ∩ territory ∩ period (with exclusion reasons for the other 11); computes per-tier payout, coverage limit used (burn %), refill fees, and 3-party net positions (Primary Insurer, All Reinsurers, Lion Re).	Deterministic `Decimal` engine; 100% exact match against hand-checked worked examples.
2. "How was this calculated?"	Step-by-step audit drawer per contract: regional peak wind/MMI, vulnerability curve damage %, insured value × damage % = gross loss, and exact layer formula with citations.	Every intermediate number is inspectable and reproducible.
3. Cited Wording Flags & Endorsement Detective	Flags clauses that bite on *this* event (e.g., T-004 storm-surge exclusion on Jebi's 329 cm Osaka tide) and transcribes scanned image endorsements (T-004 p.7: 168h replaces 72h).	2-stage verifier loop; unverified flags visibly withheld (`n verified · k withheld`); scanned pages badged `transcribed` .
4. Clause Courtroom (Dispute Resolution)	Parallel ADK agents (`cedent_counsel` ‖ `reinsurer_counsel`) argue opposing readings from verified quotes (with dynamic `COVER` clause retrieval & `...` fragment matching); engine prices both readings; Arbiter summarises without deciding.	Issue-spotting only; human analyst records the decision with an audit timestamp.
5. Bounded What-if & Grounded Q&A	Interactive storm-window scenario pricing (clearly labelled HYPOTHETICAL on Jebi's 9-hour window) and live follow-up Q&A over contract clauses screened by Google Model Armor.	Typed, bounded tool schema; Model Armor prompt injection & leakage filtering; 0 untraceable figures.
6. Instant Replay, 9-Stop Tour & Word Export	6-second zero-cost replay of a recorded real run, live SSE streaming (~9–12 s), colour-blind-safe themes, keyboard navigation, and Word (`.docx`) & Markdown memos.	Every export carries the methodology version, verification counts, and mandatory disclaimers.

Process flow diagram or Use-case diagram

Workflow 1: Live Event Analysis Graph (ADK 2.11 `Workflow`)

1. event_intel
Loads NOAA IBTrACS / USGS ShakeMap footprint, event window (`start` , `end` , `duration_h`) & JMA facts

▼

2. exposure_analyst (NO LLM — Pure Python Engine)
Matches 24 treaties → regional gross loss → Cat XL / QS waterfall → emits `result` SSE frame in ~0.7–1.0 s

▼

3. wording_checker (Gemini 3.5 Flash-Lite)
Firestore 768-d vector search per treaty (`treaty_id` pre-filter) → candidate flags with page, clause & quote

▼

4. verifier ⇄ reviser (Bounded 1-Retry Loop)
Stage 1: deterministic normalised + `...` fragment quote match · Stage 2: Gemini 3.8 Flash semantic support check → 1 revision retry on failure → withhold if still unverified

▼

5. reporter (Gemini 3.5 Flash-Lite + Python Number Gate)
Drafts executive memo from verified facts; Python checks every number against engine output (0 untraceable figures)

Workflow 2: On-Demand Clause Courtroom (Parallel Fan-Out / Join)

1. brief
Selects top verified HIGH/MEDIUM exclusion flag + loads Clause 2 (Business Covered) & exclusion clauses

▼

2a. cedent_counsel
Gemini 3.5 Flash-Lite · argues insurer reading + excluded share (0–50%)

2b. reinsurer_counsel
Gemini 3.5 Flash-Lite · argues reinsurer reading + excluded share (0–50%)

▼ JoinNode + Quote & Support Check (with 1 revision retry per side)

3. price (Deterministic Python Scenario Engine)
Prices both verified readings through the treaty layers; computes `money_at_stake` per party by subtraction

▼

4. arbiter (Gemini 3.8 Flash) → Human Analyst Decision
Summarises both arguments without deciding; analyst records position (`cedent` / `reinsurer` / `unresolved`)

Wireframes/Mock diagrams of the proposed solution (optional)

Five-Step Progressive Analyst Interface Wireframe

Sticky Summary Bar (Always Visible)
Gross Claims ($733.04M) · Reinsurers Pay ($490.92M) · Lion Re Net ($70.36M) · Verified Flags · Elapsed · Methodology 2026.1

Step 1 · Event
Track / ShakeMap + JMA facts + Instant Replay (6 s) or Run Live (~9–12 s)

Step 2 · Who's Hit
Custom SVG choropleth map of 34 prefectures + accessible exposure table

Step 3 · What It Costs
3-party bar + 13 responding treaties + expandable "How was this calculated?" layer formulas

Step 4 · Clauses That Matter
Verified wording flags + Side-by-side PDF/Clause Drawer + Parallel Clause Courtroom + What-if

Step 5 · Decide & Export
Analyst position selector + figure-checked Executive Report + Word (`.docx`) / Markdown export

Trust Architecture: 2-Stage Citation Verification Funnel

Candidate Wording Flag / Counsel Argument
Model cites `treaty_id` , `page` , `clause_no` , and verbatim `quote`

▼ Stage 1: Deterministic Python Quote Match

Word-Aligned Normalised & Ellipsis (`...`) Matcher
Checks quote against stored Firestore chunk; tags scanned pages as `transcribed` (40/40 corrupted quotes caught in unit suite)

▼ Stage 2: Semantic Support Check (Gemini 3.8 Flash)

Entailment & Event-Window Verification
Verifies quote + event window (`start` , `end` , `duration_h`) + JMA facts directly support the flag

▼ Fail → 1 Revision Retry with Feedback → Still Fails?

Visibly Withheld (Never Shown as a Finding)
UI displays `n of n displayed citations verified · k withheld` with exact reason

Architecture diagram of the proposed solution

Analyst / Judge Browser (React 18 + TypeScript + Custom SVG Choropleth)
Same-origin HTTPS · REST + POST Server-Sent Events (`fetch` ReadableStream with 15 s heartbeat)

▼

Google Cloud Run — `catsight-api` (us-central1)
Single container serves React UI + FastAPI SSE API · Google ADK 2.11 `Workflow` graphs · Deterministic Python `Decimal` Loss Engine · Dedicated least-privilege SA (`catsight-run`) · `max-instances 3` · Per-IP hourly & daily cost caps

▼

Gemini Enterprise Agent Platform
(formerly Vertex AI, global)
- Gemini 3.5 Flash-Lite
- Gemini 3.8 Flash
- gemini-embedding-001

Cloud Firestore (Native)
us-central1
- Treaty book & events
- 252 clause vectors (768-d)
- Saved runs & positions

Cloud Storage
us-central1
- Synthetic treaty PDFs
- Stamped endorsements

▼

Cloud Logging · Cloud Trace (OpenTelemetry) · Cloud Monitoring · Log Analytics SQL
Structured JSON logs per stage · Guardrail metrics (`citation_withheld` , `counsel_dropped` , `unsupported_figures`) · Live Service Dashboard

Key Architectural Decisions

- **Single-origin deployment:** React UI and FastAPI SSE stream from one Cloud Run service — zero CORS issues or proxy buffering timeouts.
- **Zero API keys in code or env:** Application Default Credentials via dedicated `catsight-run` IAM service account.
- **Pre-embedded standard queries:** Ship in `query_embeddings.jsonl` so cold starts make zero embedding calls during an analysis; interactive `/ask` queries bypass batch rate-limit sleeps.
- **Spoof-resistant rate limiting:** Reads right-most hop of `X-Forwarded-For` appended by Cloud Run's load balancer; falls back gracefully to recorded replay if cap is reached.

Technologies to be used in the solution

Technology / Service	Component	Why it was chosen & role in CatSight
Gemini 3.5 Flash-Lite Gemini Enterprise Agent Platform (Vertex AI)	Agent generation & OCR	Fast, low-cost structured output (`response_schema` , `thinking_level=LOW`) for wording checks, parallel Courtroom counsel, report prose, and scanned-page transcription.
Gemini 3.8 Flash Gemini Enterprise Agent Platform (Vertex AI)	Verification & Arbiter	Stronger reasoning model used as the semantic support judge in the Verifier Loop, Counsel Quote Checker, and neutral Courtroom Arbiter.
Google ADK 2.11 `google-genai` + ADK Workflow	Multi-agent orchestration	Graph `Workflow` with bounded retry loops (`verifier ⇄ reviser`), parallel fan-out/fan-in (`JoinNode` for blind counsel), and OpenTelemetry Cloud Trace spans.
Cloud Firestore + Vector Search `gemini-embedding-001` (768-d)	Operational & vector store	Stores the 24-treaty book, 252 clause vectors with equality pre-filtering on `treaty_id` (186/186 governing clauses retrieved), ingestion audit trail, and runs.
Google Cloud Run & Cloud Storage	Serverless runtime & PDFs	Scale-to-zero container serving FastAPI SSE + React 18 UI; GCS stores treaty PDF wordings.
Cloud Logging, Trace & Monitoring	Production observability	Log-based metrics + Log Analytics SQL query (`observability_runs.sql`) powering the Cloud Monitoring dashboard.
Python 3.12 · FastAPI · React 18 · Custom SVG	Core engine & frontend	Deterministic `Decimal` loss engine (874 offline tests + 13 opt-in live tests), python-docx memo generator, accessible TypeScript UI.

Estimated implementation cost (optional)

$0.010
p50 model cost per live analysis ($0.008–$0.013)

$0.004
p50 model cost per Clause Courtroom run

$0.000
Instant Replay cost (cached real SSE stream)

$0.003
Follow-up question (from token counts)

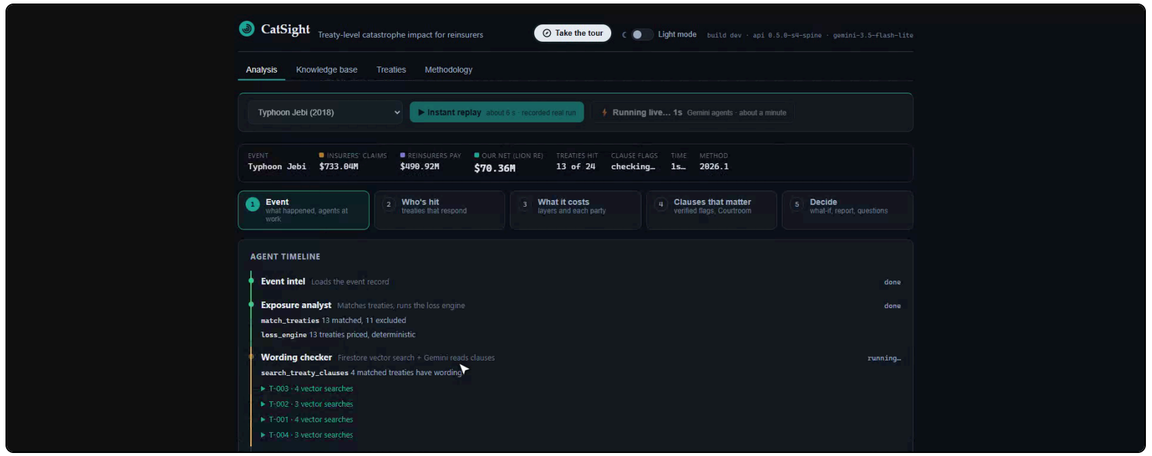
Measured Cloud & Token Breakdown

Workload	Model / Service	Measured Cost
One-time ingestion (8 PDF wordings, 252 chunks)	`gemini-embedding-001` + `3.5-flash-lite`	< **$0.08 total** (57 s runtime)
Live event analysis (13 responding treaties)	`3.5-flash-lite` + `3.8-flash` verifier	**$0.0094–$0.0102 p50**
Clause Courtroom (parallel counsel + arbiter)	2× `3.5-flash-lite` + `3.8-flash`	**$0.0040 p50**
Cloud Run + Firestore (scale-to-zero default)	`min-instances 0`, `max-instances 3`	**~$0 idle** · pennies per active hour

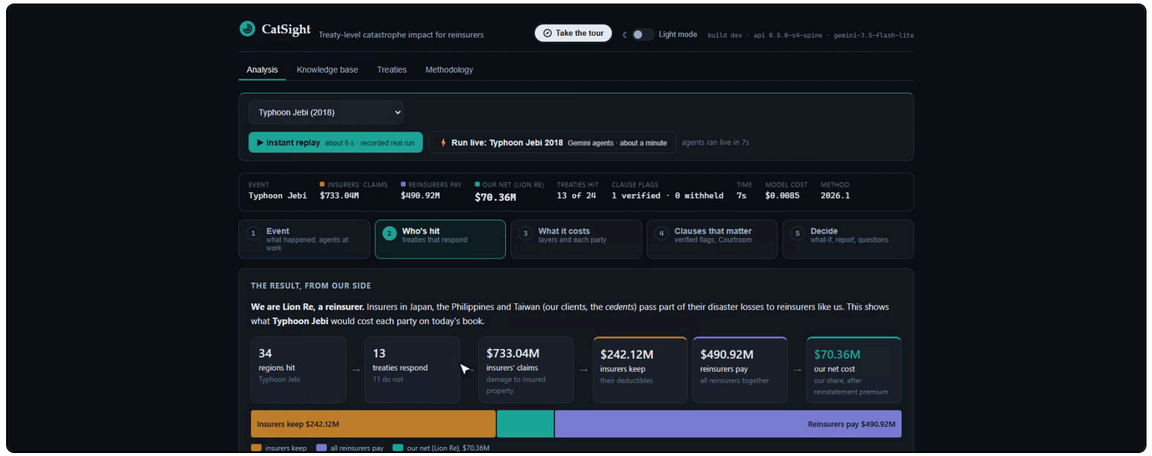
Annual Run-Rate vs. Manual Analyst Effort

- **Mid-size reinsurer workload:** 8 major events + 20 smaller checks/year (≈ 500 live analyses & Courtroom runs/year) → **~$7/year in Gemini token cost** + ~$25–$40/month Cloud Run/Firestore when active.
- **Estimated manual effort saved:** 30–60 analyst-hours per event reduced to 3.5–8 hours of review → **165–675 analyst-hours/year** (est. $12k–$100k/year in specialist time; labelled *simulated estimate*).
- **Built-in cost guards:** 20 live runs/hour per visitor, 300/day service cap, automatic fallback to Instant Replay, and Cloud Billing budget alert.

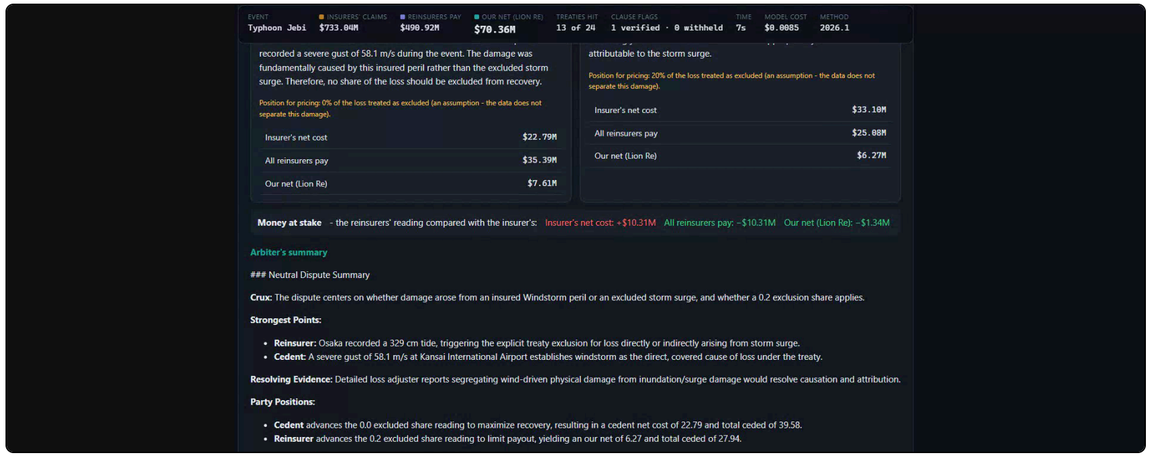
Snapshots of the prototype



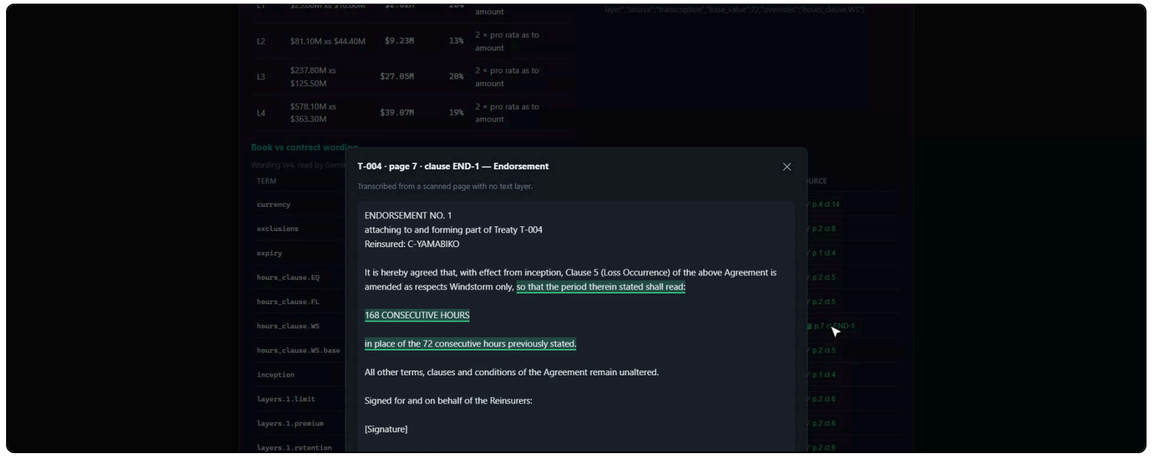
1. Live Agent Streaming: ADK 2.11 stages stream over SSE; deterministic loss figures arrive in ~0.7–1.0 s while Gemini retrieves & verifies treaty clauses.



2. Three-Party Loss Waterfall: 13 of 24 treaties respond on Typhoon Jebi ($733.04M claims → $242.12M insurer retention, $490.92M reinsurers, $70.36M Lion Re net).



3. Clause Courtroom (Jebi T-004 Storm-Surge Exclusion): Parallel counsel argue 0% vs 20% surge share; engine prices both (Ceded −$11.64M, Insurer +$10.31M, Reinsurers −$10.31M, Lion Re −$1.34M).



4. Scanned Endorsement Detective (T-004 p.7): Gemini transcribes a rasterised stamped endorsement with no text layer (168 hours replaces 72 hours) and badges it `transcribed`.

Prototype Performance report/Benchmarking

0.7–1.0 s
Time to money on screen (p50 across two 5-run live benchmarks)

9.8–12.1 s
Complete live analysis with verified flags (p50; range 8.0–31.2 s)

99.5%
RAG treaty-term extraction (185/186 fields in 57 s vs ~25 min whole-doc)

874
Automated pytest tests (874 offline tests + 13 opt-in live tests; CI on GitHub)

Benchmark / Evaluation (`docs/RESULTS.md`)	Measured Result	Methodology & Honest Caveat
Loss engine vs. hand-checked worked example	**100% exact match**	`Decimal` half-up; verified on every Cat XL layer, reinstatement, RIP, and Quota Share formula.
Governing clause retrieval & citation audit	**186/186 retrieved · 185/186 verified**	Firestore 768-d vector search; 184 text-layer verified + 1 scanned endorsement transcribed; 1 unverified withheld.
Clause Courtroom live on Cloud Run	**3 of 3 verified live (100% rate)**	Both counsel verified on first attempt; $10.3M to $26.0M money at stake priced live; 0 unsupported figures (build 6fbb81c).
Corrupted quote rejection (Verifier unit suite)	**40/40 rejected (0 false withholds)**	Normalised + word-aligned `...` ellipsis fragment matcher + Gemini 3.8 Flash semantic check (offline stubs suite).
Untraceable numbers in AI-written reports	**0 across all live runs**	Deterministic Python post-check compares every figure in reporter/arbiter prose against engine output.
Honest limitations & unmeasured items	Extraction prompt was tuned on the 8 synthetic wordings (99.5% is an upper bound; blind set is next). Manual effort savings (30–60h → 3.5–8h) are labelled *simulated estimates* pending a timed practitioner study.	

Additional Details/Future Development (if any)

The Money Moment: Why Per-Party Pricing Matters

1. Live Jebi Courtroom (T-004 Storm-Surge Exclusion, p.2 cl.8):
At a 20% surge share (recorded live run), total ceded drops by **−$11.64M** ($39.58M → $27.94M), the insurer's net cost rises by **+$10.31M**, all reinsurers save **−$10.31M**, and Lion Re saves **−$1.34M** (at 30% share: Lion Re saves **−$2.44M**).

2. Worked Example Hours-Clause Asymmetry (What-if):
On a $54.4M loss across L1 (20 xs 10, Lion Re 25%) and L2 (30 xs 30, Lion Re 10%), splitting 1 event into 2 ($35.0M + $19.4M) cuts market ceded recoveries by **−22.5% (−$10.0M)**, but **increases Lion Re's net loss by +7.4% (+$0.51M, $6.82M → $7.33M)** because the split shifts burden down into Layer 1 where Lion Re's share is concentrated.

Team Techno Crackers — Roles

• **Venket Raj S (Lead):** Architecture, Cloud Run / Firestore / IAM deployment & observability
• **Malik Chandra Pandian:** Reinsurance domain modelling, `Decimal` loss engine & worked examples
• **Arun Veerappan:** ADK 2.11 workflows, RAG extraction, citation verifier & Clause Courtroom
• **Nagaraj:** React 18 + TypeScript + custom SVG choropleth UI, 5-step analyst workflow, tour & exports

Future Development Roadmap (Designed / Investigated)

- **Real multi-shock hours-clause dispute (2024 Hualien Earthquake, Taiwan):** Investigated in repo — its real aftershock cluster ~20 days later creates a genuine second occurrence worth ~$11–15M per treaty under the hours clause.
- **Calibrated per-country vulnerability curves:** Adding insurance-penetration & insured-loss wind curves for the Philippines (bringing Typhoon Haiyan into the live selector).
- **Vendor hazard footprint adapter:** Plugging licensed Moody's RMS or Verisk footprints behind our typed hazard interface.
- **Authenticated treaty upload & BigQuery exposure:** Scaling from 24 treaties to enterprise treaty packs.
- **Live USGS / GDACS Pub/Sub intake & A2A/MCP endpoint:** Automated first-view triggers on disaster alerts and broker-agent interoperability.

Provide links to your:

- **GitHub Public Repository**
  https://github.com/venketraj/catsight-ai-buildercup-2026
  Apache-2.0 licence · 874 offline tests + 13 opt-in live tests · GitHub Actions CI · Full architecture & assumptions in `docs/`.

- **Demo Video Link (3 Minutes)**
  https://drive.google.com/file/d/14Yv-A-KtVksOgRxOxafZcEKFBUvFumQi/view?usp=sharing
  3-minute walkthrough (2m 45s) of the reinsurance problem, live ~15 s Typhoon Jebi analysis, scanned endorsement transcription, and Clause Courtroom.

- **Final Product Link**
  https://catsight-api-342162443546.us-central1.run.app
  Live on Google Cloud Run (us-central1) · No sign-up required · Click **Instant replay** (~6 s) or **Run live** (~15 s) · 9-stop guided tour included.

- **Submission Package Folder (All Artifacts)**
  https://drive.google.com/drive/folders/1X5cNQh4aMP-B1JIvIXyyxCL-4anxJhl4?usp=sharing
  Complete Google Drive submission folder containing pitch deck, demo video, architectural documentation, and verification reports.

Disclaimers & Data Attribution: Indicative first view on synthetic data. Not a catastrophe-model output. Not underwriting or claims advice. Issue-spotting only — not legal advice. Jebi, Hagibis, Haiyan and Noto were real events in which people died. Portfolio, insurers and treaty wordings are 100% synthetic; hazard data is public (NOAA IBTrACS v04r01, USGS ShakeMap, Natural Earth).

Google Cloud H2S

AI Builder Cup

Build in JAPAC.
Build for the world.

Thank you

